King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam I Term 223

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Which one of the following differential equations is Exact?

(a) $(x + \tan^{-1} y)\,dx + (y + \tan^{-1} x)\,dy = 0$

(b) $(6xy - y^3)\,dx + (x^3 - 3xy^2)\,dy = 0$

(c) $\left(\dfrac{y}{x} + y\right)dx + \left(\dfrac{x}{y} + x\right)dy = 0$

(d) $(y\cos x + \ln y)\,dx + \left(\sin x + \dfrac{x}{y}\right)dy = 0$

(e) $(xe^y + y^2)\,dx + (ye^x + 2xy)\,dy = 0$

2. Find all values of k for which the vectors of $\mathbb{R}^3$

$$u = (3, -4, 5), \quad v = (1, 1, 0), \quad w = (k, k+1, -k)$$

are linearly independent.

(a) $k \neq \dfrac{1}{2}$

(b) $k \neq 3$

(c) $k \neq 1$

(d) $k \neq \dfrac{3}{7}$

(e) $k \neq \dfrac{5}{7}$

3. Find a general solution of the differential equation

$$\sqrt{x}\,\frac{dy}{dx} = 2\cos^2 y.$$

(a) $\tan\sqrt{y} = 2\sqrt{x} + C$

(b) $\tan y^2 = \sqrt{x} + C$

(c) $y = \tan^{-1}\left(\dfrac{2}{\sqrt{x}} + C\right)$

(d) $\cos^2 y = 3\sqrt{x} + C$

(e) $y = \tan^{-1}(4\sqrt{x} + C)$

4. Solve the initial-value problem

$$y' - y\tan x = \sin x, \qquad y\left(\frac{\pi}{4}\right) = \sqrt{2}$$

(assume $x \in \left(0, \frac{\pi}{2}\right)$).

(a) $y = -\cos x + \dfrac{3}{2}\sec x$

(b) $y = -\dfrac{1}{2}\cos x \cdot \cot x + \dfrac{5}{4}\csc x$

(c) $y = \dfrac{1}{2}\cos x + \dfrac{3}{4}\sec x$

(d) $y = \dfrac{1}{2}\cos x \cdot \cot x + \dfrac{3}{4}\csc x$

(e) $y = -\dfrac{1}{2}\cos x + \dfrac{5}{4}\sec x$

5. Let $u = (4, 1)$, $v = (-2, -1)$, $w = (-3, 1)$ be vectors in $\mathbb{R}^2$. If $w = au + bv$, then $a + b =$

(a) 1

(b) 0

(c) $-\dfrac{3}{2}$

(d) -6

(e) -4

6. A pizza is removed from a freezer at $-25°$C, and left to rest at room temperature which is 20°C. After 30 minutes, the temperature of the pizza is 5°C. What is the temperature of the pizza after 60 minutes?

(a) 20

(b) 18

(c) 15

(d) 22

(e) 25

7. By making a suitable substitution, the differential equation

$$x\frac{dy}{dx} + 6y = 3xy^{4/3}$$

can be transformed into the differential equation

(a) $\dfrac{dv}{dx} - 2v = x$

(b) $\dfrac{dv}{dx} - \dfrac{2}{x}v = -2$

(c) $\dfrac{dv}{dx} - \dfrac{3}{x}v = -1$

(d) $\dfrac{dv}{dx} - \dfrac{6}{x}v = -x$

(e) $\dfrac{dv}{dx} - \dfrac{2}{x}v = -1$

8. Find a general solution of the exact differential equation

$$(e^x + ye^{xy})\,dx + (e^y + xe^{yx})\,dy = 0.$$

(a) $ye^x + xe^y = C$

(b) $e^{xy} - e^x - e^y = C$

(c) $e^x + e^y + xye^{xy} = C$

(d) $e^x + e^y + e^{xy} = C$

(e) $e^x + e^y + 2e^{xy} = C$

9. Solve the differential equation

$$\frac{dy}{dx} = (2x + y)^2 - 2.$$

(a) $y = -\dfrac{2x^2 + 2Cx + 1}{x + C}$

(b) $y = -\dfrac{x^2 + Cx + 2}{x + C}$

(c) $y = -x \ln|x + C|$

(d) $y = \dfrac{2x^2 + 2x + 1}{x + 1} + C$

(e) $y = \dfrac{2x^2 + x + 1}{x + C}$

10. A general solution of the differential equation $y' = 3\sqrt{xy}$ is $y = \left(x^{3/2} + C\right)^2$. Which one of the following statements is True?

(a) $y = 0$ is a singular solution

(b) there is no singular solution

(c) $y = \left(x^{3/2} - 1\right)^2$ is a singular solution

(d) $y = 1$ is a singular solution

(e) $y = x^3$ is a singular solution

11. Find all values of r for which $y = xe^{rx}$ is a solution of the differential equation

$$y'' + 4y' + 4y = 0.$$

(a) $r = -2$

(b) $r = 0$ and $r = 1$

(c) $r = -2$ and $r = 2$

(d) $r = -3$

(e) $r = -1$ and $r = 1$

12. Which one of the following statements is True about the following subsets of $\mathbb{R}^4$.

$$U = \{(x_1, x_2, x_3, x_4) : x_2x_3 = 0\}$$
$$V = \{(x_1, x_2, x_3, x_4) : x_1 = 1\}$$
$$W = \{(x_1, x_2, x_3, x_4) : x_4 = 2x_1\}$$

(a) W is a subspace of $\mathbb{R}^4$

(b) U is a subspace of $\mathbb{R}^4$

(c) V is a subspace of $\mathbb{R}^4$

(d) U, V and W are subspaces of $\mathbb{R}^4$

(e) none of U, V, and W is a subspace of $\mathbb{R}^4$

13. Find a general solution of the differential equation

$$y'' = 2(y')^2$$

(assume $y' > 0$).

(a) $y = B + \dfrac{1}{2}\ln|2x + A|$

(b) $y = B + 2\ln|2x + A|$

(c) $y = B - 2\ln|2x + A|$

(d) $y = B - \ln|2x + A|$

(e) $y = B - \dfrac{1}{2}\ln|2x + A|$

14. A particle is moving in a straight line with acceleration

$$a(t) = 12(t+1)^2,$$

an initial position $x(0) = 4$, and an initial velocity $v(0) = 5$. Find $x(1)$ (the position of the particle at $t = 1$).

(a) 25

(b) 35

(c) 20

(d) 15

(e) 30

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	Only $M_y = N_x = \cos x + \frac{1}{y}$ holds for $(y\cos x + \ln y)dx + (\sin x + \frac{x}{y})dy$.
2	**(e)**	$\det = 5 - 7k \neq 0$, so $k \neq \frac{5}{7}$.
3	**(e)**	Separable: $\sec^2 y\, dy = 2x^{-1/2}dx$, $\tan y = 4\sqrt{x} + C$.
4	**(e)**	$\rho = \cos x$: $(y\cos x)' = \sin x \cos x$, $y = -\frac{1}{2}\cos x + \frac{5}{4}\sec x$.
5	**(d)**	$4a - 2b = -3$, $a - b = 1$ gives $a = -\frac{5}{2}$, $b = -\frac{7}{2}$, so $a + b = -6$.
6	**(c)**	$T = 20 - 45e^{-kt}$, $e^{-30k} = \frac{1}{3}$, so $T(60) = 20 - \frac{45}{9} = 15$.
7	**(e)**	$n = \frac{4}{3}$, $v = y^{-1/3}$: $v' - \frac{2}{x}v = -1$.
8	**(d)**	$F = e^x + e^{xy} + g(y)$, $g' = e^y$: $e^x + e^y + e^{xy} = C$.
9	**(a)**	$u = 2x + y$: $u' = u^2$, $u = -\frac{1}{x+C}$, so $y = -\frac{2x^2+2Cx+1}{x+C}$.
10	**(a)**	$y = 0$ solves the ODE but no C gives it, so it is singular.
11	**(a)**	Substitute: $(r+2)^2 x + 2(r+2) = 0$ for all x, so $r = -2$.
12	**(a)**	U and V fail closure; W $(x_4 - 2x_1 = 0)$ is a subspace.
13	**(e)**	$u = y'$: $u' = 2u^2$, $u = -\frac{1}{2x+A}$, so $y = B - \frac{1}{2}\ln\lvert 2x + A\rvert$.
14	**(c)**	$v = 4(t+1)^3 + 1$, $x = (t+1)^4 + t + 3$, so $x(1) = 20$.